EVERY OBJECT OF THE BIG PRO-ETALE SITE IS COVERED BY SPECTRA OF W-CONTRACTIBLE RINGS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file proetale.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the construction of the covering through faithfully flat ind-etale extensions by w-contractible rings, the explicit cardinality estimate showing that the pieces are objects of the big pro-etale site, and the conclusion of weak contractibility.

Editorial note. The label lemma-get-many-weakly-contractible names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source proetale.tex –label lemma-get-many-weakly-contractible –start 3086 –end 3127 –local-id 3586 returned RESULT=OK (L3586 free, occupancy FREE, public interval clear). Candidate taken from sources/stacks-proetale/candidates-halfA.json.

Editorial note. The proof opens by addressing set-theoretic concerns explicitly ('For those readers who do not care about set-theoretical issues this lemma is a trivial consequence ... Here are the details.'); the details that follow are the argument, so the statement is counted.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Existence of faithfully flat ind-etale extensions by w-contractible rings. Same chapter, the proposition producing a faithfully flat ind-etale ring map from a ring to a w-contractible ring. Supplies the maps A_i -> D_i with D_i w-contractible used to build the covering.
- Equivalence between w-contractibility and weak contractibility. Same chapter, the lemma identifying w-contractible affine schemes with weakly contractible objects of the big pro-etale site. Supplies the conclusion that the spectra of the w-contractible rings are weakly contractible.
- The construction of the big pro-etale site and its size function. Same chapter, the definition of the big pro-etale site including the size bound, and the lemma describing which schemes enter the constructed category. Supply the criterion by which Spec(D_i) is isomorphic to an object of the big pro-etale site, namely the bound on the cardinality of D_i.
- Estimates for the size of affine schemes and of w-contractible rings. Sets, the lemmas bounding the size of affine schemes and of schemes of finite type, and Same chapter, the remark on the size of a w-contractible ring.

Date: stacks-project:proetale.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Supply the estimate of the cardinality of D_i in terms of the size of T, which is within the chosen bound.

Source reference. Original reference 098G: sources/stacks-proetale/pinned-original-source.tex, source lines 2465–2465. Prerequisite pointer invoked at proof line(s) 3089; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 098H: sources/stacks-proetale/pinned-original-source.tex, source lines 3027–3027. Prerequisite pointer invoked at proof line(s) 3099; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0983: sources/stacks-proetale/pinned-original-source.tex, source lines 2135–2135. Prerequisite pointer invoked at proof line(s) 3100; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0984: sources/stacks-proetale/pinned-original-source.tex, source lines 2171–2171. Prerequisite pointer invoked at proof line(s) 3124; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 000P: sources/stacks-chow/tags.txt, source lines 34–34. Prerequisite pointer invoked at proof line(s) 3121; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 04T7: sources/stacks-chow/tags.txt, source lines 6245–6245. Prerequisite pointer invoked at proof line(s) 3122; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 000J: sources/stacks-chow/tags.txt, source lines 28–28. Prerequisite pointer invoked at proof line(s) 3119; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 000X: sources/stacks-chow/tags.txt, source lines 42–42. Prerequisite pointer invoked at proof line(s) 3113; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $Sch_{pro\text{-}étale}$ be a big pro-étale site as in Definition 098G. For every object T of $Sch_{pro\text{-}étale}$ there exists a covering $\{T_i \to T\}$ in $Sch_{pro\text{-}étale}$ with each T_i affine and the spectrum of a w-contractible ring. In particular, T_i is weakly contractible in $Sch_{pro\text{-}étale}$.*

Proof. For those readers who do not care about set-theoretical issues this lemma is a trivial consequence of Lemma 098H and Proposition 0983. Here are the details. Choose an affine open covering $T = \bigcup U_i$. Write $U_i = \mathrm{Spec}(A_i)$. Choose faithfully flat, ind-étale ring maps $A_i \to D_i$ such that D_i is w-contractible as in Proposition 0983. The family of morphisms $\{\mathrm{Spec}(D_i) \to T\}$ is a pro-étale covering. If we can show that $\mathrm{Spec}(D_i)$ is isomorphic to an object, say T_i, of $Sch_{pro\text{-}étale}$, then

$\{T_i \to T\}$ will be combinatorially equivalent to a covering of $Sch_{pro\text{-}étale}$ by the construction of $Sch_{pro\text{-}étale}$ in Definition 098G and more precisely the application of Sets, Lemma 000X in the last step. To prove $\mathrm{Spec}(D_i)$ is isomorphic to an object of $Sch_{pro\text{-}étale}$, it suffices to prove that $|D_i| \leq Bound(\mathrm{size}(T))$ by the construction of $Sch_{pro\text{-}étale}$ in Definition 098G and more precisely the application of Sets, Lemma 000J in step (3). Since $|A_i| \leq \mathrm{size}(U_i) \leq \mathrm{size}(T)$ by Sets, Lemmas 000P and 04T7 we get $|D_i| \leq \kappa^{2^{2^{2^\kappa}}}$ where $\kappa = \mathrm{size}(T)$ by Remark 0984. Thus by our choice of the function $Bound$ in Definition 098G we win. □